

Decision Tree¹

The analysis of complex decisions with significant uncertainty can be confusing because 1) the consequence that will result from selecting any specified decision alternative cannot be predicted with certainty, 2) there are often a large number of different factors that must be taken into account when making the decision, 3) it may be useful to consider the possibility of reducing the uncertainty in the decision by collecting additional information, and 4) a decision maker's attitude toward risk-taking can impact the relative desirability of different alternatives. This tutorial reviews decision tree analysis procedures for addressing such complexities.

To illustrate the analysis approach, a decision tree is used in the following example to help make a decision.

Example 1.1: Product decision

To absorb some short-term excess production capacity at its Arizona plant, Special Instrument Products is considering small manufacturing run for either of two new products, a temperature sensor, or a pressure sensor. The market for each product is known if the products can be successfully developed. However, there is some chance that it will not be possible to develop them successfully.

Revenue of \$1,000,000 would be realized from selling the temperature sensor, and revenue of \$400,000 would be realized from selling the pressure sensor. Both of these amounts are net of production cost but do not include development cost. If development is unsuccessful for a product, then there will be no sales, and the development cost will be totally lost. The development cost would be \$100,000 for the temperature sensor and \$10,000 for the pressure sensor.

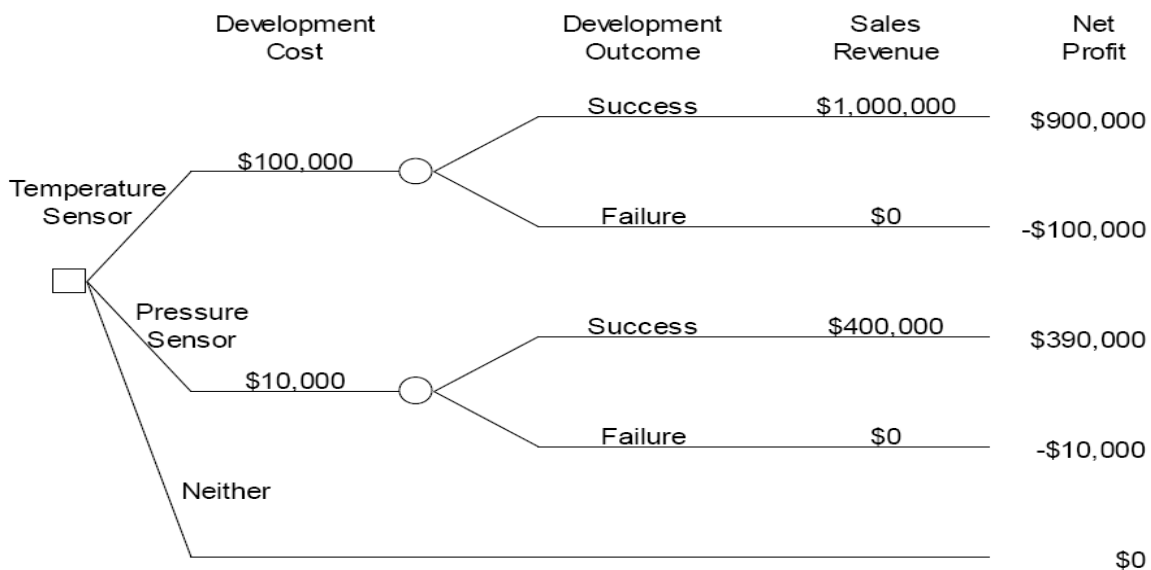


Figure 1: Special Instrument Products decision

¹ Developed based on Decision Tree Primer by Craig W. Kirkwood.

Decision tree notation

A diagram of a decision, as illustrated in the above figure, is called a decision tree. This diagram is read from left to right. The leftmost node in a decision tree is called the root node. In the above figure, this is a small square called a decision node. The branches emanating to the right from a decision node represent the set of available decision alternatives. One, and only one of these alternatives, can be selected. The small circles in the tree are called chance nodes. The right end of each path through the tree is called an endpoint, and each endpoint represents the final outcome of following a path from the root node of the decision tree to that endpoint.

Question: Which, if either, of these products, should Special Instrument Products attempt to develop?

You may be thinking that the decision about which alternative is preferred depends on the probabilities that development will be successful for the temperature or pressure sensors. This is indeed the case. Suppose that experts estimate that the probability of development success is 0.5 for the temperature sensor and 0.8 for the pressure sensor. While only the probability of success is specified for each development effort, the probability of failure is determined by the rules of probability since the probabilities of development success, and development failure must add up to one.

In the decision tree, probabilities are shown in parentheses on each branch of a chance node.

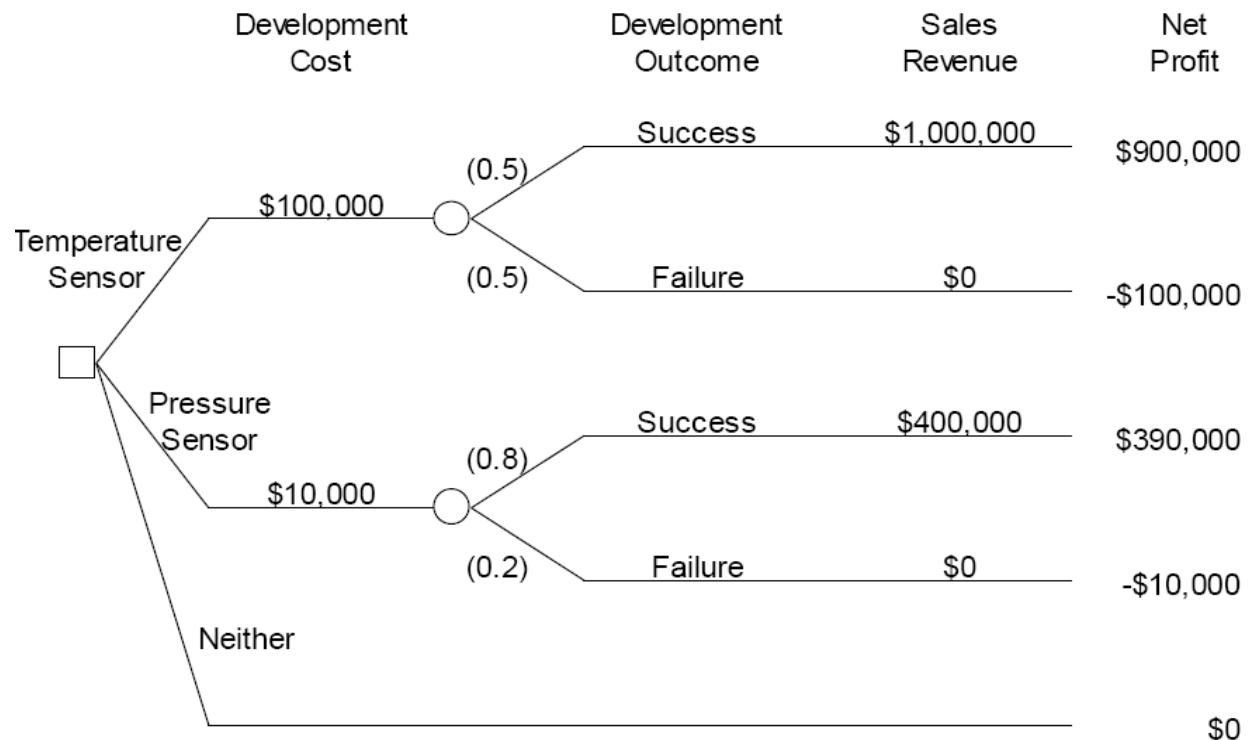


Figure 2: Special Instrument Products decision tree with probabilities

A study of figure 2 shows that knowing the probabilities will not by itself always make the best alternative in a decision immediately clear.

Expected Value

To decide which alternative to select in a decision problem, we need a decision criterion; that is, a rule for making a decision. Expected value is a criterion for making a decision that takes into account both the possible outcomes for each decision alternative and the probability that each outcome will occur.

The expected value for an uncertain alternative is calculated by multiplying each possible outcome of the uncertain alternative by its probability and summing the results. The expected value decision criterion selects the alternative that has the best-expected value. In situations involving profits where “more is better,” the alternative with the highest expected value is best, and in situations involving costs, where “less is better,” the alternative with the lowest expected value is best. Expected value is often a good measure of the value of an alternative since, over the long run, this is the average amount that you expect to make from selecting the alternative.

To illustrate the concept of expected value, we consider a simpler decision with lower stakes than the Special Instrument Products decision.

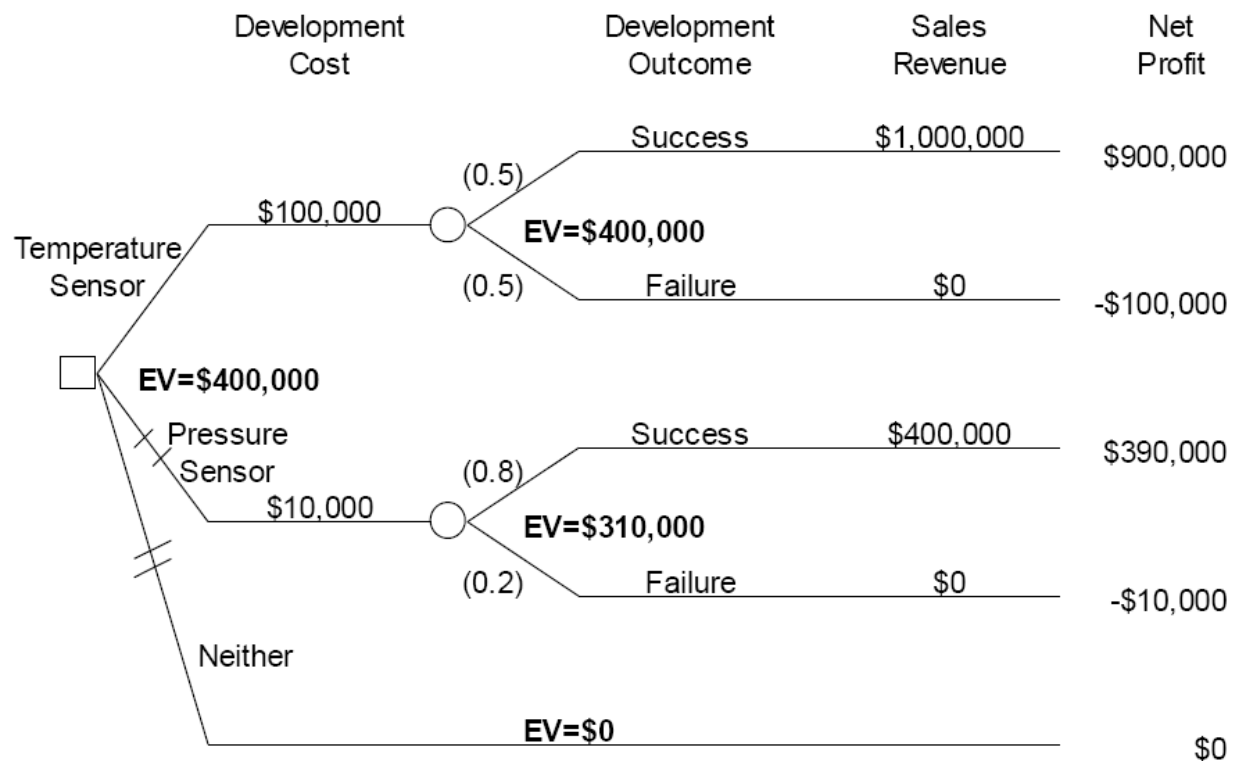


Figure 3: Special Instrument Products decision tree, with expected values

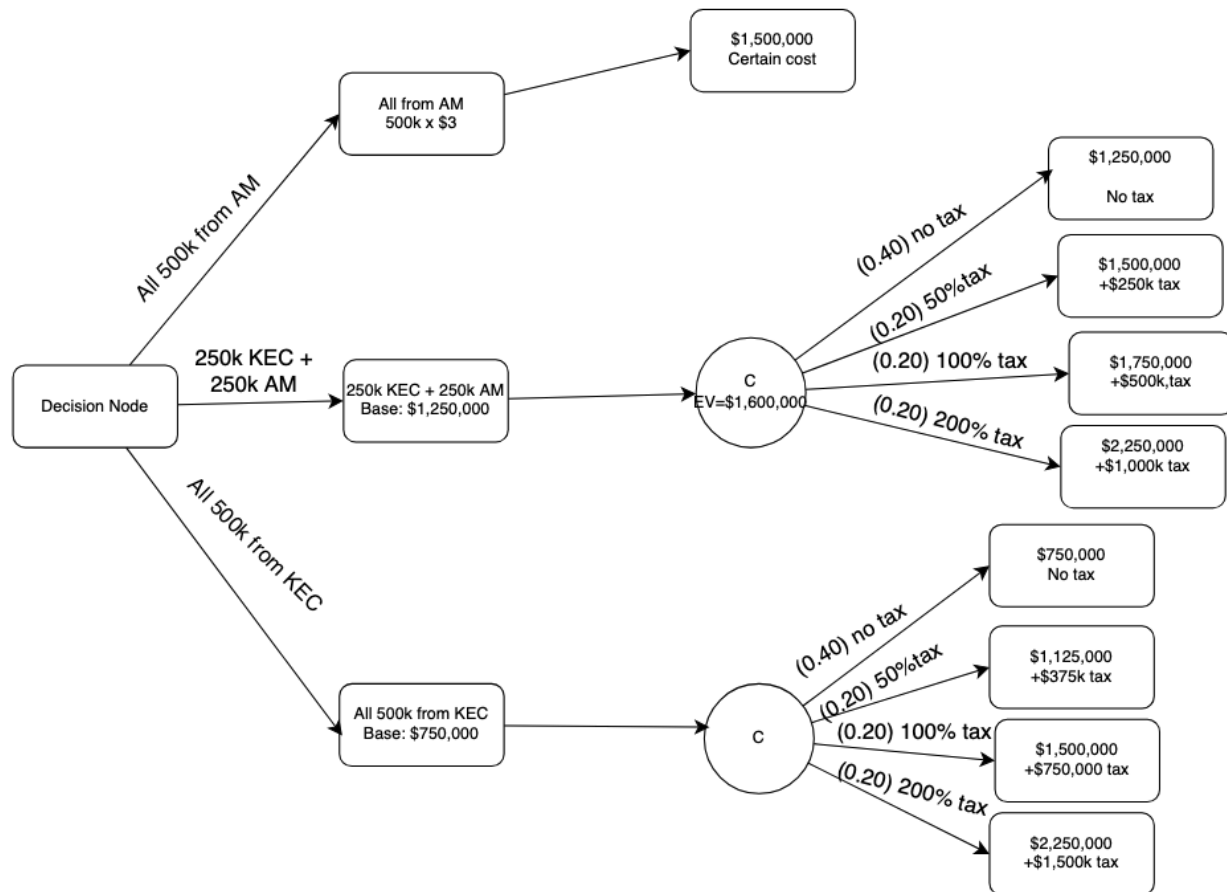
Figure 3 illustrates some additional notation that is often used in decision trees. The branches representing the two alternatives that are less preferred are shown with crosshatching (//) on their branches. The expected value for each chance node is designated by “EV.” Finally, the expected value at the decision node on the left is shown as equal to the expected value of the selected alternative. Thus, the alternative with the highest expected value is developing the temperature sensor, and if the expected value criterion is applied, then the temperature sensor should be developed.

Exercise

Kezo Systems has agreed to supply 500,000 PC FAX systems to Tarja Stores in 90 days at a fixed price. A key component in the FAX systems is a programmable array logic integrated circuit chip ("PAL chip"), one of which is required in each FAX system. Kezo has bought these chips in the past from an American chip manufacturer AM Chips. However, Kezo has been approached by a Korean manufacturer, KEC Electronics, which is offering a lower price on the chips. This offer is open for only 10 days, and Kezo must decide whether to buy some or all of the PAL chips from KEC. Any chips that Kezo does not buy from KEC will be bought from AM. AM Chips will sell PAL chips to Kezo for \$3.00 per chip in any quantity. KEC will accept orders only in multiples of 250,000 PAL chips and is offering to sell the chips for \$2.00 per chip for 250,000 chips, and for \$1.50 per chip in quantities of 500,000 or more chips. However, the situation is complicated by a dumping charge that has been filed by AM Chips against KEC. If this charge is upheld by the U.S government, then the KEC chips will be subject to an anti-dumping tax. This case will not be resolved until after the point in time when Kezo must make the purchase decision. If Kezo buys the KEC chips, these will not be shipped until after the anti-dumping tax would go into effect, and the chips would be subject to the tax. Under the terms offered by KEC, Kezo would have to pay any anti-dumping tax that is imposed. Kezo believes there is a 60% chance the anti-dumping tax will be imposed. If it is imposed, then it is equally likely that the tax will be 50%, 100%, or 200% of the sale price for each PAL chip.

- i. Draw a decision tree for this decision.

Below



- ii. Using expected value as the decision criterion, determine Kezo's preferred ordering alternative for the PAL chips.

Exected Value Calculations

Alternative 1: All from AM

Cost = $500,000 \times 3,000 = \$1,500,000$

Alternative 2: 250k KEC + 250k AM

Base cost = $(250,000 \times 2.00) + (250,000 \times 3.00) = \$500,000 + \$750,000 = \$1,250,000$

Outcome	Probability	TotalCost
Notax	0.40	\$1,250,000
50%tax(+\$250k)	0.20	\$1,500,000
100%tax(+\$500k)	0.20	\$1,750,000
200%tax(+\$1,000k)	0.20	\$2,250,000

$$EV = (0.40 \times \$1,250,000) + (0.20 \times \$1,500,000) + (0.20 \times \$1,750,000) + (0.20 \times \$2,250,000)$$

$$EV = \$500,000 + \$300,000 + \$350,000 + \$450,000$$

Alternative 3: All 500k from KEC

$$\text{Base cost} = 500,000 \times \$1.50 = \$750,000$$

Outcome	Probability	Total Cost
No tax	0.40	\$750,000
50% tax (+\$375k)	0.20	\$1,125,000
100% tax (+\$750k)	0.20	\$1,500,000
200% tax (+\$1,500k)	0.20	\$2,250,000

$$EV = (0.40 \times \$750,000) + (0.20 \times \$1,125,000) + (0.20 \times \$1,500,000) + (0.20 \times \$2,250,000)$$

$$EV = \$300,000 + \$225,000 + \$300,000 + \$450,000 = \mathbf{\$1,275,000}$$

Using the expected value criterion, Kezo should order all 500,000 chips from KEC Electronics. This alternative has the lowest expected cost at \$1,275,000 saving \$225,000 compared to buying everything from AM. Although ordering entirely from KEC carries risk, the 200% tax scenario would result in a \$2,250,000 cost, the attractive base price of \$1.50/chip combined with the 40% chance of no tax makes this the optimal choice under the expected value criterion.